

The data (14 samples)					
RID	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31..40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31..40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31..40	medium	no	excellent	yes
13	31..40	high	yes	fair	yes
14	>40	medium	no	excellent	no

Class counts

Yes: 9

No: 5

Total: 14

Formulas

Entropy of a node:

$$\text{Info}(D) = - \sum p_i \cdot \log_2(p_i)$$

Weighted entropy after splitting on attribute *A*:

$$\text{Info}_A(D) = \sum (|D_j| / |D|) \cdot \text{Info}(D_j)$$

Information Gain:

$$\text{Gain}(A) = \text{Info}(D) - \text{Info}_A(D) \rightarrow \text{pick the largest Gain}$$

Step 1 - Entropy of the whole dataset

$$\text{Info}(D) = -(9/14)\log_2(9/14) - (5/14)\log_2(5/14)$$

$$\text{Info}(D) = 0.940$$

Step 2 - Gain for each attribute (root candidates)

age (values: <=30 / 31-40 / >40)

value	samples	yes/no	Entropy
<=30 (youth)	5	2 / 3	0.971
31...40 (middle-aged)	4	4 / 0	0.000
>40 (senior)	5	3 / 2	0.971

$$\text{Info}_{\text{age}}(D) = (5/14)(0.971) + (4/14)(0) + (5/14)(0.971)$$

$$\text{Info}_{\text{age}}(D) = 0.694$$

$$\text{Gain}(\text{age}) = 0.940 - 0.694$$

$$\text{Gain}(\text{age}) = 0.246$$

income (values: high / medium / low)

value	samples	yes/no	Entropy
high	4	2 / 2	1.000
medium	6	4 / 2	0.918
low	4	3 / 1	0.811

$$\text{Info}_{\text{income}}(D) = (4/14)(1.000) + (6/14)(0.918) + (4/14)(0.811)$$

$$\text{Info}_{\text{income}}(D) = 0.911$$

$$\text{Gain}(\text{income}) = 0.940 - 0.911$$

$$\text{Gain}(\text{income}) = 0.029$$

students (values: yes/no)

value	samples	yes/no	Entropy
yes	7	6 / 1	0.592
no	7	3 / 4	0.985

$$\text{Info}_{\text{student}}(D) = (7/14)(0.592) + (7/14)(0.985)$$

$$\text{Info}_{\text{student}}(D) = 0.788$$

$$\text{Gain}(\text{student}) = 0.940 - 0.788$$

$$\text{Gain}(\text{student}) = 0.151$$

credit_rating (values: fair / excellent)

value	samples	yes/no	Entropy
fair	8	6 / 2	0.811
excellent	6	3 / 3	1.000

$$\text{Info}_{\text{credit}}(D) = (8/14)(0.811) + (6/14)(1.000)$$

$$\text{Info}_{\text{credit}}(D) = 0.892$$

$$\text{Gain}(\text{credit_rating}) = 0.940 - 0.892$$

$$\text{Gain}(\text{credit_rating}) = 0.048$$

Root decision

attribute	gain
age	0.246
student	0.151
credit_rating	0.048
income	0.029

Step 3 - Recurse on each branch of age

branch age = 31..40 (middle-aged)

All samples are yes (entropy 0) / leaf = YES

branch age = <30 (youth)

5 samples (2 yes, 3 no), entropy 0.971

split on student	samples	yes/no	Entropy
yes	2 (RID 9, 11)	2 / 0	0.000
no	3 (RID 1, 2, 8)	0 / 3	0.000

$$\text{Info}_{\text{student}} = 0 \rightarrow \text{Gain } 0.971$$

- student = no -> Leaf = NO

- student = yes -> Leaf = YES

branch age = >40 (senior)

5 samples (3 yes, 2 no), entropy 0.971

split credit_rating	samples	yes/no	Entropy
fair	3 (RID 4, 5, 10)	3 / 0	0.000
excellent	2 (RID 6, 14)	0 / 2	0.000

$$\text{Info}_{\text{credit}} = 0 \rightarrow \text{Gain } 0.971$$

- credit_rating = fair -> Leaf = YES

- credit_rating = excellent -> Leaf = NO

Final decision Tree

